

Laporan Implementasi Data Warehouse

ETL Data Sensor IoT menggunakan Python + SQLite

Identitas Mahasiswa

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- Kelas: C
- Mata Kuliah: Data Warehouse

Ringkasan

- ETL berhasil dijalankan dari notebook secara berurutan.
- Data sumber CSV berhasil dimuat ke star schema SQLite.
- Validasi 4 tabel utama berhasil.
- Minimal 3 query analitik dijalankan dan diinterpretasikan.

Skema Data Warehouse (star schema)

- fact_sensor_readings (fact table)
- dim_sensors (dimension)
- dim_locations (dimension)
- dim_time (dimension)

Hasil validasi jumlah record:

- dim_sensors: 10
- dim_locations: 5
- dim_time: 1440
- fact_sensor_readings: 14400

Bukti Instalasi dan Proses ETL

Berikut screenshot bukti instalasi dependency serta bukti eksekusi notebook ETL. Notebook status eksekusi: 27/27 code cell sudah dieksekusi.

Bukti verifikasi dependency environment

```
Dependency environment verification
Interpreter: /workspaces/.venv/bin/python

pandas: available (version 3.0.6)
numpy: available (version 2.4.6)
sqlalchemy: available (version 2.1.1)
matplotlib: available (version 3.11.2)
jupyter: available (version installed)
```

Gambar 1. Bukti dependency environment

Screenshot proses ETL (eksekusi notebook)

Diambil otomatis dari environment dan artefak proyek.

```
Notebook: hands_on_data_warehouse.ipynb
Total code cells: 27
Executed code cells: 27
Unexecuted code cells: 0
Status: SUCCESS
```

Gambar 2. Bukti eksekusi notebook ETL

Validasi Hasil Loading dan Alur ETL

Alur ETL:

- 1) Extract: membaca CSV processed_sensor_data_20250930_092513.csv.
- 2) Transform: konversi timestamp, standarisasi lokasi, deduplikasi, handling missing value.
- 3) Load: insert ke tabel dimensi dan fact pada SQLite warehouse.

Screenshot di bawah menunjukkan validasi jumlah record untuk 4 tabel yang diminta.

Screenshot validasi tabel warehouse

Diambil otomatis dari environment dan artefak proyek.

```
dim_sensors: 10
dim_locations: 5
dim_time: 1440
fact_sensor_readings: 14400
```

```
Schema tables:
- dim_locations
- dim_sensors
- dim_time
- fact_sensor_readings
- sqlite_sequence
```

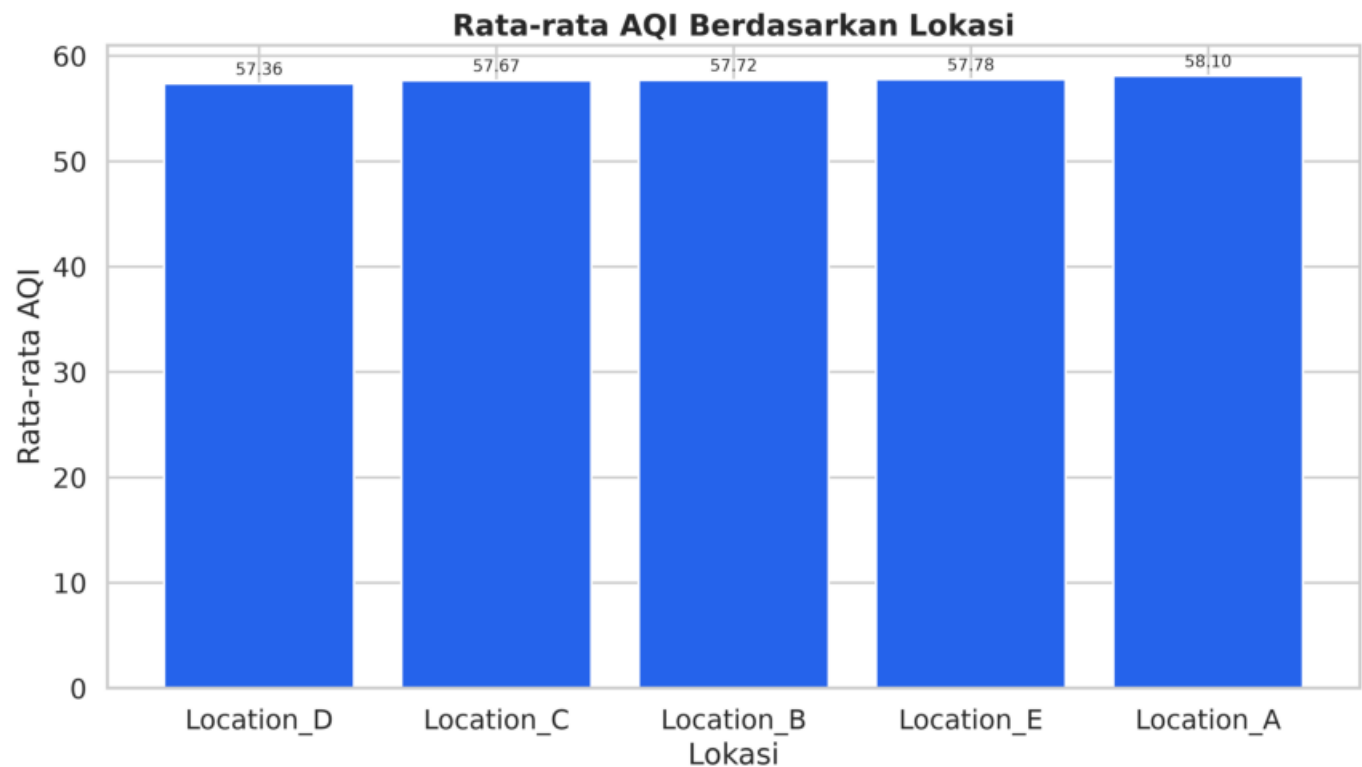
Gambar 3. Validasi dim_sensors, dim_locations, dim_time, fact_sensor_readings

Query Analitik 1 - Kualitas Udara per Lokasi

SQL Query:
SELECT l.location_name, ROUND(AVG(f.air_quality_aqi),2) AS avg_aqi, COUNT(*) AS n
FROM fact_sensor_readings f JOIN dim_locations l ON f.location_key=l.location_key
GROUP BY l.location_name ORDER BY avg_aqi ASC;

Hasil utama (5 baris): [('Location_D', 57.36, 2880), ('Location_C', 57.67, 2880), ('Location_B', 57.72, 2880), ('Location_E', 57.78, 2880), ('Location_A', 58.1, 2880)]

Insight: lokasi dengan AQI rata-rata terendah (lebih baik) adalah Location_D.



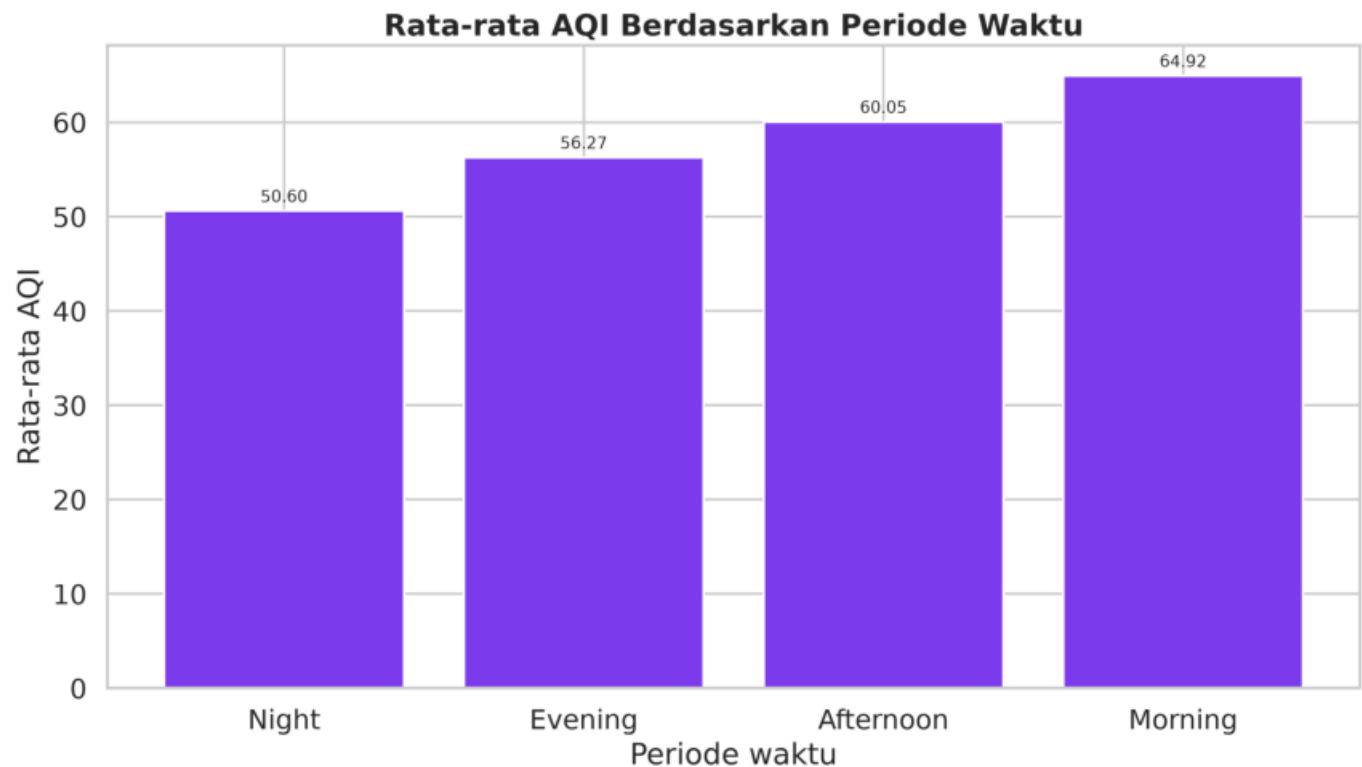
Gambar 4. Visualisasi rata-rata AQI per lokasi

Query Analitik 2 - Pola AQI per Periode Waktu

SQL Query:
SELECT t.time_period, ROUND(AVG(f.air_quality_aqi),2) AS avg_aqi, COUNT(*) AS n
FROM fact_sensor_readings f JOIN dim_time t ON f.time_key=t.time_key
GROUP BY t.time_period ORDER BY avg_aqi ASC;

Hasil: [('Night', 50.6, 4200), ('Evening', 56.27, 3000), ('Afternoon', 60.05, 3600), ('Morning', 64.92, 3600)]

Insight: periode Night memiliki AQI rata-rata terendah.



Gambar 5. Visualisasi AQI rata-rata berdasarkan periode waktu

Query Analitik 3 - Aktivitas Sensor dan Kendala

SQL Query:

```
SELECT s.sensor_id, COUNT(*) AS n_readings
FROM fact_sensor_readings f JOIN dim_sensors s ON f.sensor_key=s.sensor_key
GROUP BY s.sensor_id ORDER BY n_readings DESC, s.sensor_id ASC;
```

Hasil utama: [('SENSOR_001', 1440), ('SENSOR_002', 1440), ('SENSOR_003', 1440), ('SENSOR_004', 1440), ('SENSOR_005', 1440), ('SENSOR_006', 1440), ('SENSOR_007', 1440), ('SENSOR_008', 1440), ('SENSOR_009', 1440), ('SENSOR_010', 1440)]

Insight: semua sensor memiliki jumlah pembacaan setara (1.440), sehingga tidak ada satu sensor yang dominan.

Screenshot hasil 3 query analitik

Diambil otomatis dari environment dan artefak proyek.

```
[Query 1] Rata-rata AQI per lokasi (terbaik 5)
('Location_D', 57.36, 2880)
('Location_C', 57.67, 2880)
('Location_B', 57.72, 2880)
('Location_E', 57.78, 2880)
('Location_A', 58.1, 2880)

[Query 2] Rata-rata AQI per periode waktu
('Night', 50.6, 4200)
('Evening', 56.27, 3000)
('Afternoon', 60.05, 3600)
('Morning', 64.92, 3600)

[Query 3] Jumlah pembacaan per sensor (top 10)
('SENSOR_001', 1440)
('SENSOR_002', 1440)
('SENSOR_003', 1440)
('SENSOR_004', 1440)
('SENSOR_005', 1440)
('SENSOR_006', 1440)
('SENSOR_007', 1440)
('SENSOR_008', 1440)
('SENSOR_009', 1440)
('SENSOR_010', 1440)
```

Gambar 6. Screenshot hasil 3 query analitik

Kendala dan solusi:

- Kendala: sempat muncul IndentationError pada data_loader.py.
- Solusi: pembersihan karakter asing pada file dan validasi ulang syntax.
- Kendala: interpretasi loaded records pada run ulang.
- Solusi: pemisahan metrik records in warehouse vs loaded this run.